STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 1A- Basic Concepts in Probability 1

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Session Outline

1. Random Experiment
    - Random Variable
2. Definition of Probability
3. Axioms and Laws of Probability
    - Axioms of Probability
    - Properties of Probability
4. Expectation of Random Variable

Session Overview

The purpose of this session is to equip students with the basic knowledge and skills in probability. In this session, students will be introduced to basic concepts in probability such as random events, random variables and expectation of random variables.

Learning Outcomes

- At the end of the session, student will be able to:
  1. Define and distinguish between random events and random variable.
  2. State examples of random events and random variables.
  3. Define the expectation of a random variable.
  4. Calculate an expectation of a given probability function.

Random Experiment

- An experiment is a process of observation.
- A random experiment is any process of observation or activity in which the results can not be predicted with certainty.
- Alternatively, an experiment or random experiment is any specified set of actions, where the results cannot be predicted with certainty.
- Each repetition of an experiment is called a trial.
- Typical examples of random experiments are:
  a. the roll of a die,
  b. the toss of a coin,
  c. selecting a ball at random from a box containing 3 balls marked “a”, “b”, “c”.

Definition of Some Terminologies

- In describing an experiment, the following terms are employed; **Outcomes, Events and Sample Space**.
- *An outcome* is a result of an experiment.
  Example: When a fair coin is tossed once, the result is either a **Head** or **Tail**, which are known as outcome.
- *An event* is any collection of outcomes, and a simple event is an event with only one possible outcome.
  Example: In tossing a fair die once, the possible outcomes are $\{1, 2, 3, 4, 5, 6\}$. The collection of one or more of these possible outcomes is an event, i.e. $\{1, 2, 3\}$ or $\{4\}$.
- The *sample space, S* of a given experiment is a set that contains all possible outcomes of the experiment.
  Example: In the example of tossing a fair die, the sample space is $\{1, 2, 3, 4, 5, 6\}$.

Definition of Some Terminologies

- A *sample point*, **s** is defined as one possible outcome of a random experiment. In the example of tossing a fair coin, ”Head” or ”Tail” is the sample point.
- Note that a set with one sample point, is often referred as an elementary event.
- A set A is called a subset of B, denoted by $A \subset B$ if every element of A is also an element of B. Any subset of the sample space is called an event.
  Example: If $S = \{1, 2, 3, 4, 5, 6\}$ and set $A = \{2, 3\}$, set $B = \{1, 2, 3, 4\}$, then we say $A \subset B$.
- The sample space, S is the subset of itself, that is $S \subset S$. Since S is the set of all possible outcomes of the experiment, it is often referred to as certain event.

Example 1

Find the sample space for an experiment of tossing a coin repeatedly and counting the number of tosses required until the first head appears.

Solution :
It is easy to realize that all possible outcomes for this experiment are the terms of the sequence $1, 2, 3, \ldots$.
Thus the sample space is given by; $S = \{1, 2, 3, \ldots\}$.

Examples Cont'd.

Example 2

Consider an experiment of drawing two cards at random from a bag containing four cards marked with the integers $1, 2, 3, 4$.

a. Find the sample space S_1 of the experiment if the first card drawn is not replaced. Let the first number indicates the number of the first card drawn.
b. Find the sample space S_2 of the experiment if the first card drawn is replaced before the second one is drawn.

Solution

Solution (a)

The sample space S_1 contains 12 ordered pairs (i, j), where $i \neq j$, $1 \leq i \leq 4$ and $1 \leq j \leq 4$; the first number indicates the first card drawn. Thus:

$$S_1 = \begin{pmatrix} (1,2) & (1,3) & (1,4) \\ (2,1) & (2,3) & (2,4) \\ (3,1) & (3,2) & (3,4) \\ (4,1) & (4,2) & (4,3) \end{pmatrix}$$

Solution (b)

The sample space S_2 contains 16 ordered pairs (i, j), where $1 \leq i \leq 4$ and $1 \leq j \leq 4$, the first number indicates the first card drawn. Thus:

$$S_2 = \begin{pmatrix} (1,1) & (1,2) & (1,3) & (1,4) \\ (2,1) & (2,2) & (2,3) & (2,4) \\ (3,1) & (3,2) & (3,3) & (3,4) \\ (4,1) & (4,2) & (4,3) & (4,4) \end{pmatrix}$$

Example 3
Find the sample space for the experiment of measuring (in hours) the life time of a transistor.

Solution (3): Clearly all possible outcomes are all positive real numbers, i.e. the life time can take all values on a positive number line. Thus the sample space S is given by; $S = \{0 \leq X < \infty\}$.

- **Note:** It is useful to distinguish between two types of sample spaces: **Discrete** and **Continuous**.

- A sample space is discrete if it contains a finite or countable infinite set of possible outcomes. In this case the elements in the sample space can be listed or the list does not terminate as in example 1.

- A sample space is continuous if it contains an interval (either finite or infinite) of real numbers as seen in Example 3 above.

Random Variable: Definition

- A random variable, usually written X, is a variable whose possible values are numerical outcomes of a random phenomenon.
- Alternatively: A random variable is a variable whose value is unknown or a function that assigns values to each of an experiment's outcomes.
- Random variables are often designated by letters (X, Y or Z)and can be classified as:

  discrete, which are variables that have specific values, or

  continuous, which are variables that can have any values within a continuous range.

Examples of Random Variables

1. A typical example of a random variable is the outcome of a coin toss. If the random variable, Y, is the number of heads we obtained from tossing two coins, then Y could be 0, 1, or 2. This means that we could have no heads, one head, or both heads on a two-coin toss.

2. Examples of discrete random variables include the number of children in a family, the Friday night attendance at a cinema, the number of patients in a doctor's surgery, the number of defective light bulbs in a box of ten.

3. Continuous random variables are usually measurements. Examples include height, weight, the amount of sugar in an orange, the time required to run a mile.

Definition of Probability: Introduction

- Chance and the assessment of risk play a part in everyone's life.
- As a loan officer you may want to evaluate the probability that a customer defaults.
- As an investment officer you are interested in finding the likelihood of an investment results in a loss.
- Probability has found a wide range of business applications such as: calculation of risk in the banking and insurance industries, quality control and market research.
- Most events in life are unpredictable, i.e. events in life are non-deterministic.
- If the world is treated as stochastic then we can measure the chances of any particular outcome happening in a given situation.

Probability Definition

- Probability is the measure of the likelihood of the occurrence of an event.
- The probability of an event can be thought of as the relative frequency of the event.
- The probability of an outcome can also be interpreted as a subjective probability, or degree of belief that the outcome will occur.
- Probabilities are numbers between 0 and 1.
- An event with probability 0 has no chance of occurring, and an event with probability 1 is certain to occur.
- Other definitions of Probability are Relative Frequency and Classical definitions.

Relative Frequency Definition

- The probability of an outcome is interpreted as the limiting value of the proportion of times the outcome occurs in n repetitions of the random experiment as n increases beyond all bounds.
- For example if we assign a probability of 0.4 to the outcome that there is a defective item in a production batch, we might interpret this assignment as implying that, if we analyse many items in the production batch, approximately 40% of them will be defective.
- This example illustrates a relative frequency interpretation of probability.

Relative Frequency Definition

$$P(A) = \lim_{n\to\infty} \frac{n(A)}{n},$$

- where $\frac{n(A)}{n}$ is called the relative frequency of event A.
- Stated differently, the relative frequency can be stated as the ratio $\frac{n(A)}{n}$, for the probability of A, if n is a large number.
- When the likelihood assumption is not valid, then the relative frequency method can be used.
- The relative frequency technique, lead one to conduct an experiment n times and record the outcomes.
- The probability of event A is assigned by $P(A) = \frac{f_A}{n}$, where f_A denotes the number of experimental outcomes that satisfy event A.

Worked Example

The table below shows the frequency distribution of grades in Statistics class.

Marks	60	75	78	80	83	86	88	90	92	96	98	99
Freq	1	3	4	5	7	11	13	5	7	4	2	1

The relative frequency for mark 90 is computed as:

$$\frac{5}{63} = 0.079$$

Also the relative frequency for obtaining a mark less than or equal to 83 is

$$\frac{1+3+4+5+7}{63} = \frac{20}{63} = 0.31$$

Using the relative frequency definition, we can therefore estimate probabilities. Thus the probability of obtaining a mark 90 is 0.079 and that obtaining a mark less than or equal to 83 is 0.31.

Classical Definition (Equally Likelihood Model)

- In an experiment for which the sample space is S and the outcomes are equally likely (that is, each outcome has the same chance of occurring), then the probability of an event A occurring is given by the formula

$$P(A) = \frac{\text{number of outcome in A}}{\text{total number of outcomes}}$$

$$= \frac{\text{number of ways that } A \text{ can occur}}{\text{number of ways the sample space } S \text{ can occur}}$$

$$= \frac{n(A)}{n(S)}$$

- In other words, $P(A)$ is the fraction of the total number of outcomes that cause the event A to occur.

Example 1

- Suppose there are 5 balls in a box, 3 balls are red and 2 are black. The box is shaken, and a ball selected without looking. What is the probability that this ball is red?

- **Solution :**

- The event that a ball selected is red consists of 3 outcomes out of 5 possible outcomes in all. According to the basic probability formula, the probability of selecting a red ball is

$$P(\text{red ball}) = \frac{3}{5}.$$

Example 2

In a group of 400 college students, there are 170 science majors and 230 non-science majors. Of the science majors, 55 are women, and there are 120 women non-science majors. One student is selected at random from the group. Find the probability that the selected student is:

1. A Science major
2. A Woman
3. Neither a woman nor a science major

Solution :

	Science Majors	Non -Science Majors	Total
Women	55	120	175
Men	115	110	225
Total	170	230	400

Solution Cont'd

- Let A denotes the event that a student is a science major, i.e. $n(A) = 170$; also let S denotes the total sample size, $n(S) = 400$.
- Ans 1: There are 170 science majors out of 400 students, so $P(Science major) = P(A) = \frac{n(A)}{n(S)} = \frac{170}{400}$.
- Let W represents the event a student is a woman, $n(W) = 175$, since there are 175 women out of 400 students altogether, we have

  $P(Woman) = P(W) = \frac{n(W)}{n(S)} = \frac{175}{400}$
- There are 110 students who are neither women nor science majors so $P(\text{neither woman nor science major}) = \frac{110}{400}$

Axioms of Probability

Let S be a finite sample space, and A be an event in S. Then in the axiomatic definition, the probability $P(A)$ of the event A is a real number assigned to A which satisfies the axioms:

- **Axiom 1:** $0 \leq P(A) \leq 1$.
  Probability is a number between 0 and 1 (inclusive)
- **Axiom 2:** $P(S) = 1$ and $P(\phi) = 0$.
  In performing an experiment, we assume that the result will be one of the simple outcomes. In other words, we assume that the event S will occur.
- **Axiom 3:** (a) If A and B are mutually exclusive events, then $P(A \cup B) = P(A) + P(B)$.

Axioms of Probability Cont'd.

- If the sample space S is not finite (ie. infinite) then (a) must be modified as follows:

- (b) If $A_1, A_2, \ldots,$ is an infinite sequence of mutually exclusive events in S, that is $(A_i \cap B_j = \phi)$ for $i \neq j$, then

$$
\begin{aligned}
P(A_1 \cup A_2 \cup A_3 \cup \ldots) &= P\left(\bigcup_{i=1}^{\infty} A_i\right) \\
&= P(A_1) + P(A_2) + P(A_3) + \ldots \\
&= \sum_{i=1}^{\infty} P(A_i)
\end{aligned}
$$

Axioms of Probability Cont'd.

- This means for instance if $A_1, A_2, A_3, \ldots,$ are events that can not occur simultaneously, then the probability that one among them will occur is the sum of their probabilities.

- For example, the probability of either a 2 or a 6 in a roll of a die is

$$\frac{1}{6} + \frac{1}{6} = \frac{2}{6}, \quad \text{(Axiom 3)}.$$

Elementary Properties of Probability

- **Property 1:** $P(\bar{A}) = 1 - P(A)$
  This means that if the probability of occurrence of an event A is $P(A)$, then the probability of non occurrence of the event A is $P(\bar{A}) = 1 - P(A)$.
- For instance if the probability of obtaining a 6 in a roll of die is $\frac{1}{6}$ then the probability of not obtaining a 6 is $1 - (\frac{1}{6}) = \frac{5}{6}$.
- **Property 2:** $P(\phi) = 0$, the probability of null or empty set or an impossible event is zero.
- **Property 3:** The probability of a union (addition rule) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- $P(A) \leq P(B)$, if the event $A \subseteq B$.

- Finally, if A and B are mutually exclusive, then $P(A \cap B) = 0$ and the formula becomes

$$P(A \cup B) = P(A) + P(B)$$

.

- **Theorem 1:** If A, B and C are any three events, then

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C)$$

$$-P(B \cap C) + P(A \cap B \cap C)$$

The Proof is left as an Exercise.

Worked Example

Suppose events A, B and C have probabilities; $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(C) = \frac{1}{4}$. Furthermore, assume that $A \cap C = \phi$, $B \cap C = \phi$ and $P(A \cap B) = \frac{1}{6}$. Use the axioms and properties of probabilities as well as the facts concerning sets and events to find.

a. $P[(A \cap B)']$
b. $P(A \cap B')$
c. $P[(A \cup B)']$
d. $P(A' \cap B')$
e. $P(A \cup B \cup C)$

Solution

a. Applying property 1, we have
$P[(A \cap B)'] = 1 - P(A \cap B) = 1 - \frac{1}{6} = \frac{5}{6}$

b. We use the relation: $A = (A \cap B') \cup (A \cap B)$
Now, $P(A) = P[(A \cap B') \cup (A \cap B)]$
$= P(A \cap B') + P(A \cap B)$
$\implies \quad \frac{1}{2} = P(A \cap B') + \frac{1}{6}$
$\implies \quad P(A \cap B') = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$

c. $P[(A \cup B)'] = 1 - P(A \cup B)$
$= 1 - [P(A) + P(B) - P(A \cap B)]$
$= 1 - [\frac{1}{2} + \frac{1}{3} - \frac{1}{6}] = \frac{1}{3}$

d. According to De Morgan's law, we have
$P(A' \cap B') = P[(A \cup B)'] = \frac{1}{3}$

Solution Cont'd

- (e) Since $A \cap C = \phi$ and $B \cap C = \phi$, it follows that $(A \cup B) \cap C = \phi$. Also $P(A \cup C) = \frac{2}{3}$ and $P(C) = \frac{1}{4}$, then

$$P(A \cup B \cup C) = P[(A \cup B) \cup C]$$

$$= P(A \cup B) + P(C) = \frac{2}{3} + \frac{1}{4}$$

$$= \frac{11}{12}.$$

Example 2

In the toss of a fair die, find the probability that the result is even or divisible by 3.

Solution: Let A be the event that the result is even and B the event that the result is divisible by 3.
Then $A = \{2, 4, 6\}$ and $B = \{3, 6\}$, it implies that $A \cap B = \{6\}$. Now, the required probability is $P(A \cup B)$.

$$
\begin{aligned}
P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\
&= \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{2}{3}
\end{aligned}
$$

Theorem

- **Definition:** Two events A and B are said to be equally likely if $P(A) = P(B)$. Event A is said to be more likely than B if $P(A) > P(B)$.

- **Theorem 2:** If all the outcomes of a random experiment consist of n equally likely simple events, then the probability of an event E made up of k simple events is
$$P(E) = \frac{k}{n}.$$
The proof is left as an Exercise.

Trial Questions

1. A box contains 10 green balls and 8 red balls. Six balls are selected at random. Find the probability that:
   a. all 6 balls are green.
   b. that exactly 4 out of the 6 balls are red.
   c. at least one of the 6 balls is red.
   d. 3 or 4 of the 6 balls are red.
2. A bag contains two red, three green and four black balls of identical size except for colour. Three balls are drawn at random without replacement.
   a. Show that the probability that two balls have the same colour and the third is different is $\frac{55}{84}$.
   b. What is the probability that at least one is red.

Expectation of Random Variables

- An important notion in statistics is the expected value or expectation of a random variable.
- The expectation of a random variable is the long-term average of the random variable.
- For instance, imagine observing many thousands of independent random values from the random variable of interest. Take the average of these random values.
- The expectation is the value of this average as the sample size tends to infinity.

Definition of Expectation

1. If X is a random variable distributed as $p(x_i)$ or $f(x)$, then the expected value of X , which we will denote as $E(X)$, is defined as

$$E(X) = \begin{cases} \sum_{all\, x_i} x_i p(x_i), & \text{where } X \text{ is a discrete random variable} \\ \\ \int_{-\infty}^{\infty} x f(x) dx, & \text{where } X \text{ is a continuous random variable} \end{cases}$$

   - where the summation is over the entire sample space of X, i.e., over all admissible values of X and $p(x_i)$ is the probability mass function of X.
   - If the random variable is continuous, then the summation is replaced by the integration and $f(x)$ is the density function of the random variable X.

Example

Consider a class of 24 students. Suppose that 5 of them are 19 years old, 7 are 20 years old , 10 are 23 years old, and 2 are 17 years old. Let us denote by μ the mean age of the class, that is :

$$\mu = \frac{5 \times 19 + 7 \times 20 + 10 \times 23 + 2 \times 17}{24} = 20.79$$

Let X be the random variable taking the distinct ages as values, that is $X = \{x_1, x_2, x_3, x_4\}$, with $x_1 = 19$, $x_2 = 20$, $x_3 = 23$ and $x_4 = 17$, with probability law given as:
$P(X = x_1) = \frac{5}{24}, P(X = x_2) = \frac{7}{24}, P(X = x_3) = \frac{10}{24}, P(X = x_4) = \frac{2}{24}$.
We may summarize this probability law in the following table as:

x	19	20	23	17
$P(X = x)$	$\frac{5}{24}$	$\frac{7}{24}$	$\frac{10}{24}$	$\frac{2}{24}$

Example Cont'd

- Set sample space, Ω as the class of these 24 students. The following graph
  $X : \Omega \to \mathbb{R}$
  $\quad \omega \to X(\omega) =$ age of the student ω,
  defines a random variable. The probability that is used on Ω, the sample space is the frequency of occurrence.
  For each $i \in \{1, 2, 3, 4\}$, $P(X = x_i)$ is the frequency of occurrence of the age x_i in the class.
- Therefore, through these notations, we have the following expression of the mean μ:

$$\mu = \sum_{i=1}^{4} x_i P(X = x_i).$$

- Note: The Expectation will be treated later in session 3B.

THANK YOU!!!